# Bayesian Estimation in HANK*

These are some of the notes I made to guide myself through the estimation material. They simply review, in one single place, the estimation lecture, the SHADE paper part on estimation, estimation in Krusell Smith (one shock, one time series), estimation in one-asset HANK (three shocks, three time series).

For a hands on implementation, please refer to

1. sections 5 & 6 Estimation of Sticky Wage HANK with 3 Shocks.ipynb
2. section_6_new_krusell_smith.ipynb

**Page 13** has a useful **one-page summary** of the estimation algorithm.

- What shocks and models rationalize the observed macro time series?
- Macro model with $\epsilon_t \sim \mathcal{N}(\mathrm{O}, \mathrm{I})$ shocks (various policy and technology) and parameters $\theta$ (including s.s., trends, shock process parameters)
- Data observables $y_t$ (historical cosumption, investment) to get estimates + standard errors on $\theta$ and back-out the likley history of shocks $\epsilon_t$. Can do variance decompositions (what drives business cycles?), counterfactual decompositions (what if a certain shock or trend/change hadn't happened?), forecasts (what happens next?).

**Traditional State-Space Methods v.s. New Sequence-Space Jacobian Methods [with a few subtle differences I will highlight]**

- State space form: $s_t$ $n_s \times 1$ states (capital, TFP, the distribution of agents), $y_t$ observables $n_y \times 1$ (aggregate $Y$ and $C$), $\epsilon_t$ $n_\epsilon \times 1$ vector of innovations

$$\begin{aligned} s_t &= \Phi_1(\theta) s_{t-1} + \Phi_\epsilon(\theta)\epsilon_t \quad \epsilon_t \sim_{iid} \mathcal{N}(0, I) \\ y_t &= \Psi_0(\theta) + \Psi_1(\theta) s_t \end{aligned}$$

- This can be iterated forward to get an $MA(\infty)$ representation for horizon $s \to \infty$.

*Email Martin Aragoneses maragoneses@g.harard.edu if you spot mistakes or have questions. These notes are based on Ludwig Straub's wonderful lecture notes, as well as hist SHADE paper with coauthors, and their various iPython code.

$$y_t = \Psi_0 + \Psi_1 s_t = \Psi_0 + \Psi_1 \sum_{s=0}^{\infty} (\Phi_1)^s \Phi_\epsilon \epsilon_{t-s}$$

- SSJ gives this to us the $MA(\infty)$ representation directly. Recall we did this in our simple HANK settup with shocks $\epsilon_t = dG_t, dT_t$

$$Y_t = G_t + \mathcal{C}_t\left(\{Y_s - T_s\}\right), \quad dY_t = dG_t + \sum_{s=0}^{\infty} \frac{\partial \mathcal{C}_t}{\partial Z_s} \cdot (dY_s - dT_s), \quad d\mathbf{Y} = \sum_{s=0}^{\infty} \mathbf{M}^s (d\mathbf{G} - \mathbf{M} d\mathbf{T}) + d\lambda \cdot \mathbf{v}$$

- While $\mathbf{M}$ is the partial equilibrium Jacobian of $C$ to disposable income, we got that the $s$'th column of the GE Jacobian $\mathbf{G}_{Y,T} = \mathbf{M}^s \mathbf{M}$ gave us the coefficients on changes $d\mathbf{T}$.

**First Step: Analyze** model *given* parameters $\theta$, obtaining

- **IRFs** selecting shock $j$ with unit vector $e_j$,
  - SSJ gives us IRFs directly as the $h$th column of the GE Jacobian for any horizon $h$

$$\text{IRF}(j,h) \equiv \mathbb{E}\left[y_{i,t+h} \mid s_{t-1}, \epsilon_{j,t} = 1\right] - \mathbb{E}\left[y_{i,t+h} \mid s_{t-1}\right] = \Psi_1 (\Phi_1)^h \Phi_\epsilon e_j$$

- **second moments** of observables (e.g. Cov with output), given i.i.d shocks that are not directly serially correlated and have variance $I$.[1] **[different with SSJ]**
  - Fixed point on the variance covariace matrix $\Gamma_{ss}(0)$
  - With this, we can compute OLS regression coefficients (Cov/Var), annd HP filtered moments (detrending)

$$\begin{aligned}
\Gamma_{ss}(0) \equiv \quad & \mathbb{E}\left[s_0^2\right] && = \Phi_1 \Gamma_{ss}(0) \Phi_1' + \Phi_\epsilon \Phi_\epsilon' \\
\Gamma_{ss}(h) \equiv \quad & \mathbb{E}\left[s_t s_{t-h}\right] && = (\Phi_1)^h \Gamma_{ss}(0) \\
\Gamma_{yy}(h) \equiv \quad & \mathbb{E}\left[y_t y_{t-h}\right] && = \Psi_1 \Gamma_{ss}(h) \Psi_1'
\end{aligned}$$

  - SSJ gives us that shocked variables $\{dZ_t\}$ are $MA(\infty)$ in iid structural innovation vectors $\epsilon_t$. Given certainty equivalence, so is also any endogenous $\{dX_t\}$ where $M^Z(\theta)$ and $M^X(\theta)$ are simply the Jacobians, and related to eachother via the implicity function theorem giving us the GE matrix $G$ mapping shock $Z$ to outcome $X$.

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[1]

$$\begin{aligned}
V(s_0) \quad &= \quad \mathbb{E}\left[s_0 s_0'\right] = \Phi_1(\theta) \mathbb{E} s_{-1} s_{-1}' \Phi_1(\theta)' + \Phi_\epsilon(\theta) \mathbb{E} \epsilon \epsilon' \Phi_\epsilon(\theta)' \\
s_t s_{t-h}' \quad &= \left(\sum_{h=0}^{\infty} (\Phi_1)^h s_{t-h}\right) s_{t-h}' + \ldots, \quad \mathbb{E}\left[s_{t-h} s_{t-h}'\right] \to \Gamma_{ss}(0)
\end{aligned}$$

$$dZ_t = \sum_{s=0}^{\infty} M_s^Z(\theta)\epsilon_{t-s}, \quad dX_t = \sum_{s=0}^{\infty} M_s^X(\theta)\epsilon_{t-s}, \quad \mathbf{G} = -\mathbf{H}_X^{-1}\mathbf{H}_Z, \quad M^X(\theta) = G^{X,Z}(\theta)M^Z(\theta)$$

  - Given this, second moments can be simply calculated via

$$\text{Cov}\left(dX_t, dY_{t'}\right) = \sum_{s=0}^{\infty} M_s^X(\theta) \cdot M_{s+t'-t}^Y(\theta)$$

- **forecast error variance decompositions**, distribution of forecast errors at any horizon
  - What drives, "on average", movements in $y$ at different horizons? (e.g. what drives output at business cycle frequencies?)
  - Using the $MA(\infty)$ of the model with all the IRFs, we can compute, at a given horizon, the forecase error (which takes out the constant and exploits that $\mathbb{E}_t\epsilon_{t+h} = 0$.
  - $\tilde{\mathbf{e}}_{t|t-h}$ captures all movements in $y_t$ that happen after $t-h$, and we can compute it's variances and covariances (and ask questions like what accounts for the var of output in the next 3-4 years?)

$$\tilde{e}_{t|t-h} = y_t - \mathbb{E}_{t-h}\left[y_t\right] = \Psi_1 \sum_{s=0}^{h-1} \left(\Phi_1\right)^s \Phi_\epsilon \epsilon_{t-s}$$

  - The forecase error (Co)variance decomposition is the share of the total variance of observable $i$ contributed by shock $j$.

$$FEVD(i,j,h) = \frac{V_i|\text{j shock}}{V_{ii}|\text{ all shocks}} = \frac{\left[\Psi_1 \sum_{s=0}^{h-1} \left(\Phi_1\right)^s \Phi_\epsilon e_j e_j' \Phi_\epsilon' \left(\Phi_1'\right)^s \Psi_1'\right]_{ii}}{\left[\Psi_1 \sum_{s=0}^{h-1} \left(\Phi_1\right)^s \Phi_\epsilon \Phi_\epsilon' \left(\Phi_1'\right)^s \Psi_1'\right]_{ii}}$$

  - We can do this at different horizons (e.g. one period ahead, deeper into the recession, in recovery) for different aggregates (output, labor demand, investment). This is the classical exercise of Smets and Wouters.
  - SSJ gives us FEVD at any horizon $h$ directly from the Jacobians + $MA$ representation

- **historical decompositions** into likely shocks
  - we'd like to back out which shocks were likely driving historical data on $y_t$
  - we need at least $n_\epsilon \geq n_y$ enough shocks to identify the number of aggregates
  - back out the most likely shock path $\epsilon_0, \epsilon_1, ...$ that minimizes the distance

$$\min_{\{\epsilon_t\}} \sum_{s=0}^{T} \left\| y_s^{data} - \tilde{e}_{s|0}\left(\{\epsilon_t\}\right) \right\|^2$$

- Feeding in one shock at a time, can then construct a historical decomposition of the time series into shocks that explain it[2]
- What share of shocks in the 1980s, 1990s, etc. were driven by technology v.s. demand v.s. investment shocks?
- SSJ gives us historical decompositions directly from Jacobians + $MA$ representation

- **evaluating likelihood** $p\left(Y_{1:T} \mid \theta\right)$ given of observing given data $Y_{1:T}$ time series). **[different with SSJ]**
  - the likelihood function is the density $p\left(Y_{1:T}, S_{1:T} \mid \theta\right)$ based on the linearized model $y_t(s_t)$ and $s_t\left(s_{t-1}, \epsilon_t\right)$ where $Y_{1:T} \equiv \left(y_1 \dots y_T\right)$
  - since states are usually unoserved, we work with the likelihood over observables, $p\left(Y_{1:T} \mid \theta\right)$, which we need to construct iteratively (**filtering**) to avoid integrating over the distribution of states across all periods

$$p\left(Y_{1:T} \mid \theta\right) = \int p\left(Y_{1:T}, S_{1:T} \mid \theta\right) dS_{1:T}$$

  - To filter, let $p\left(s_0\right) = p\left(s_0 \mid Y_{1:0}\right)$ be the initial distribution of states (use the invariant distribution associated with the law of motion $s_t\left(s_{t-1}, \epsilon_t\right)$
  - From sequential observations of $y_t$, compute $p\left(s_t \mid Y_{1:t}\right)$ recursively as well as $p\left(y_t \mid Y_{1:t-1}\right)$ also recursively given $y_t\left(s_t\right)$
  - This is similar to what you did with stochastic simulation or for the expectations vector in the fake-news algorithm. Starting with $p\left(s_0\right)$ at the steady state, apply exogenous shocks + endogenous transitions given past states. To get the distribution for any other variable, e.g. output, simply use how it's related to $s_t$ within a period.

$$p\left(s_0|Y_{1:0}\right) = p\left(s_0\right) = \Gamma_{ss}, \quad p\left(s_t|Y_{1:t}\right) = \Phi_1 p\left(s_{t-1}|Y_{1:t}\right) + \Phi_\epsilon \mathbf{N}\left(0,1\right)$$
$$\implies p\left(y_t \mid Y_{1:t-1}\right) = \Psi_0 + \Psi_1 p\left(s_t \mid Y_{1:t-1}\right)$$

  - $p\left(y_t \mid Y_{1:t-1}\right)$ is the probability of obtaining $y_t$ given the observed time-series. Since shocks are i.i.d., the probability of obtaining the entire time series

$$p\left(Y_{1:t}\right) = \prod_{t=1}^{T} p\left(y_t \mid Y_{1:t-1}\right)$$

  - Given that $s_t$ is linear and $p\left(s_0\right)$ is Normal, we ca use the Kalman filter

[2] In practice we also back out likely shocks that hit before beginning of sample, rising penalization $\lambda$ until the data is no longer explained by $\epsilon_s$.

$$\min_{\{\epsilon_t\}_{t=-\tau}^{T} \sum_{s=0}^{T}} \left\| y_s^{data} - \tilde{e}_{s|0}\left(\{\epsilon_t\}_{t=-\tau}^{T}\right)\right\|^2 + \lambda \sum_{s=-\tau}^{T} \|\epsilon_s\|^2$$

- At any time $t-1$ we can forecast $s_t$ and $y_t$ using the density of past $s_{t-1}$ and the known transitions from $s_{t-1}$ and $s_t$ since $s_t\left(s_{t-1}, \epsilon_t\right)$ where $\epsilon_t$ has a known distribution, and $y_t(s_t)$.

$$p\left(s_t \mid Y_{1:t-1}\right)=\int p\left(s_t \mid s_{t-1}\right) p\left(s_{t-1} \mid Y_{1:t-1}\right) d s_{t-1}$$

$$\Longrightarrow \quad p\left(y_t \mid Y_{1:t-1}\right)=\int p\left(y_t \mid s_t\right) p\left(s_t \mid Y_{1:t-1}\right) d s_t$$

- Once we observe $y_t$, we can update with Bayes rule

$$p\left(s_t \mid Y_{1:t}\right)=\frac{p\left(y_t \mid s_t, Y_{1:t-1}\right) p\left(s_t \mid Y_{1:t-1}\right)}{p\left(y_t \mid Y_{1:t-1}\right)}$$

- Given $p\left(s_t \mid Y_{1:t}\right)$, we can now forecast $p\left(s_{t+1} \mid Y_{1:t}\right)$, and so on. All we need to do is keep track of $s_t \mid Y_{1:t-1}, y_t \mid Y_{1:t-1}$, and $s_t \mid Y_{1:t}$.
- With SSJ we do not need to use a Kalman filter! We stack all the covariances of observables into $\mathbf{V}(\theta)$ and the data into $Y$. Since shocks are log-normal, we can calculate the log-likelihood directly using the log of a normal density

$$\mathcal{L}(\mathbf{Y};\theta)=-\frac{1}{2}\log\det\mathbf{V}(\theta)-\frac{1}{2}\mathbf{Y}'\mathbf{V}(\theta)^{-1}\mathbf{Y}$$

- Use Cholesky decomposition of $\mathbf{V}$ to quickly calculate $\log\det\mathbf{V}$ and $\mathbf{Y}'\mathbf{V}(\theta)^{-1}\mathbf{Y}$

**Second Step:** ***Etimate*** $\theta$ using the objects above[the following steps are the same for SSJ and t]

- **IRF matching** to empirical IRFs (using VARs or local projections) $\widehat{IRF}(j,h)$
  - Compare them against the model-implied IRFs $\mathrm{IRF}(j,h;\theta)=\Psi_1(\theta)\left(\Phi_1(\theta)\right)^h\Phi_\epsilon(\theta)e_j$
  - Stack these vectors across $j,h$, weight by $\hat{V}$ diagonal matrix with estimated variances of $\widehat{IRF}(j,h)$

$$\hat{\theta}=\arg\min_\theta(\mathbf{IRF}\left(\theta\right)-\widehat{\mathbf{IRF}})'\hat{V}^{-1}(\mathbf{IRF}\left(\theta\right)-\widehat{\mathbf{IRF}})$$

- **minimum distance estimation / Simulated Method of Moments**
  - With model implied second moments $M(\theta)$ (e.g. auto-covariaces with output) annd empirical counterparts $\hat{M}$ estimated from the data with diagonal variances $\hat{V}$ of the moments

$$\widehat{\theta}=\arg\min(M(\theta)-\hat{M})'\hat{V}^{-1}(M(\theta)-\hat{M})'$$

- **maximum likelihood estimation:**
  - Frequentist ML on macro data not always super well behaved, often multiple local maxima, so can't just do

$$\widehat{\theta} = \arg\max p\left(Y_{1:T}, \theta\right)$$

- Bayesian estimation is a better option: given prior $p(\theta)$, construct the posterior mode and do ML on it instead, just like with simple ML but adding some prior.

$$p\left(\theta \mid Y_{1:T}\right) = \frac{p\left(Y_{1:T} \mid \theta\right) p(\theta)}{p\left(Y_{1:T}\right)} \propto p\left(Y_{1:T} \mid \theta\right) p(\theta)$$
$$\widehat{\theta} = \arg\max p\left(Y_{1:T} \mid \theta\right) p(\theta)$$

- Typically we do this constructing an artificial markov chain for $\theta_t$ whose stationary distribution is the posterior (aka "MCMC") with $p\left(\theta \mid Y_{1:T}\right)$. Move from $\theta_t$ to $\theta_{t+1}$ using Metropolis-Hastings

- **Advantages of SSJ**
  - estimating shock process parameters in $\theta$ (entering $M^{Z}(\theta)$) is free since we only need to calculate $\boldsymbol{G} = -\mathbf{H}_{x}^{-1}\mathbf{H}_{z}$ once, and then we can reuse $\mathrm{M}^{X} = G^{X,Z}M^{Z}$ may times without simulation
  - robust as long as we are using cross-sectional evidence to match an unchaged stationary distribution. With the same stationary distribution, Jacobians do not need to be recomputed (but assumes that the economy converges back to the old steady state).
  - for parameters that do affect the stationary distribution, estimation is significatly slower

Paper's notation.

- variables without tildes denote impulse responses, while variable with tildes denote observed time paths.

- Assume each shock $z \in \mathcal{Z}$ follows an $MA$ process

$$d\tilde{Z}_{t}^{z} = \sum_{s=0}^{\infty} dZ_{s}^{z}\epsilon_{t-s}^{z}$$

- where $\{\epsilon_{t}^{z}\}$ are mutually i.i.d. standard normal shocks. Thus, a unit innovation in $\epsilon_{t}^{z}$ starting at $t$ has an impulse response $dZ_{0}^{z}, dZ_{1}^{z}, \ldots$

- SSJ tells us how to find the IRF to any output $dX_{0}^{o}, dX_{1}^{o}, \ldots\ o \in \mathcal{O}$ in response to symultaous impulses to all shocks $z \in \mathcal{Z}$.

- Instead we are interested in the impulse of $o$ to an innovation $\epsilon_{t}^{z}$ for only a specific $z$ $dX_{0}^{o,z}, dX_{1}^{o,z}, \ldots$

- Let $\{dX_{s}^{o,z}\}_{s} \equiv d\mathbf{X}^{o,z}$ $\{dZ_{s}^{z}\}_{s} \equiv d\mathbf{Z}^{z}$. Then from our implicit function theorem, the IRF (where $d\mathbf{U} = \mathbf{G}^{U,Z}d\mathbf{Z} = -\mathbf{H}_{\mathbf{U}}^{-1}\mathbf{H}_{\mathbf{Z}}d\mathbf{Z}$ ).

$$d\mathbf{X}^{o,z} = \mathbf{G}^{o,z}d\mathbf{Z}^{z}$$

- Certainty equivalence implies that each output follows an $MA(T-1)$ process where the impulse response is $d\mathbf{X}^{o,z}$ and changes with all shocks $z \in \mathcal{Z}$ that we have in the model. Given the effect of $\epsilon^z_{t-s}$ on our outcome, which is traced by each IRF, and given the different shocks we have, we get our outcome as a combination of shocks with different IRFs.

$$d\tilde{X}^o_t = \sum_{s=0}^{T-1} \sum_{z \in \mathcal{Z}} dX^{o,z}_s \epsilon^z_{t-s}$$

- Thus, a simple $MA$ representation for the observed time path of an output $d\widetilde{\mathbf{X}}$ multiplying GE Jacobians $\mathbf{G}$ times shock impulse response vectors $d\mathbf{Z}$.

- Note: read appendix "C.8" to see how to map the state-space and sequence-space representation of the Krusell Smith model where

$$\begin{pmatrix} \mathbf{D}_{t+1} \\ K_t \\ Z_t \end{pmatrix} = \mathbf{A} \begin{pmatrix} \mathbf{D}_t \\ K_{t-1} \\ Z_{t-1} \end{pmatrix} + \mathbf{B}\epsilon_t$$

1. Assume GE IRFs $d\mathbf{X}^{o,z}$ have been computed for a given outcome $o$ for all shocks $z$, with truncation horizon $T$

2. Simulate a random sample path of $o$ is as follows

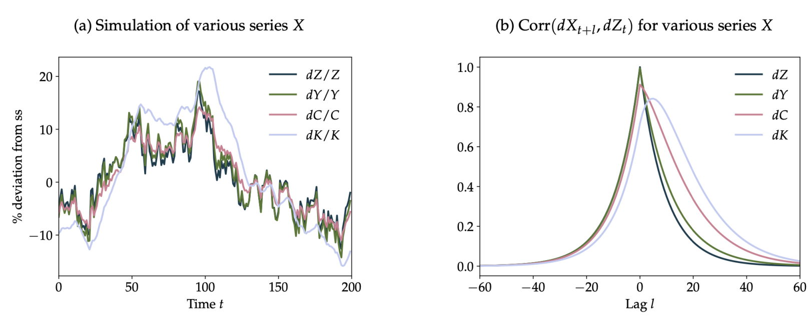
   i. **draw** paths for the shock innovations from their random i.i.d. distribution), that is, a sequence $\{\epsilon^z_t\}$ for each of the shocks $z$ upto a large truncation horizon $\mathbb{T}$

   ii. for each $t$ in the time path, evaluate how the entire shock sequence affects this aggregate

$$d\tilde{X}^o_t = \sum_{s=0}^{T-1} \sum_{z \in \mathcal{Z}} dX^{o,z}_s \epsilon^z_{t-s}$$

   iii. discard the first $T$ elements of the path $d\tilde{X}^o_t$, generating a single random sample path of outcome $o$ of length $\mathbb{T} - T$

- We can do this for any outcome $o \in \mathcal{O}$ and for any $\mathbb{T} - T$ horizonof the simulation of the time series of $d\tilde{X}^o_t$. For example, in the Krusell Smith model:



- Note this is exactly what you did for getting a single draw of the firm dyamics path in the stochastic simulation you computed in Problem Set 2. There, you were starting the simulation as an entrant with an average value of the state, and are driven each period by the stochastic path of shocks that you drew in every step of the chain. Here, instead, you are drawing a path of aggregate shocks, driving $Z_t = \rho Z_{t-1} + \varepsilon_t$ in each period, and simply using the fact that $dK = G_{K,Z} dZ$ as well as $dY = G_{Y,Z} dZ$ are know and calculate all other aggregate dynamics from there. Obviously, this gives you a single path of the dynamics of an economy, much like a single draw gave you a path for the dyamics of a single firm in the Hopenhay Rogerson simulation. The Krusell Smith model is a little different, since there's only one observable (capital), one shock (productivity). Like Hopenhayn Rogerson, you can have more than one random shock, there Poisson death and productivity. For example, the one asset HANK has three observables, three shocks.

- To compute the "expected" path of the economy driven by all of it's shocks, we can simply draw multiple simulated paths and average across them. One might be tempted to compute moments of outcomes like the variances and covariances, at various lags and leads by simulating time paths of outcomes multiple times. We have, however, a more efficient way to do this using the IRFs $d\mathbf{X}^o$.

- **Analytical second moments.**

- $d\tilde{\mathbf{X}}_t = \left(d\tilde{X}_t^o\right)_{o\in\mathcal{O}}$ = vector of stochatic processes of all $n_o$ outputs.

- $d\mathbf{X}_s = [dX_s^{o,z}]$ stacked $n_o \times n_z$ matrix of $MA$ coefficients $dX_s^{o,z}$ for each output $o$ and input shock $z$ (which we used for the IRF)

$$d\mathbf{X}^{o,z} = \mathbf{G}^{o,z} d\mathbf{Z}^z, \quad d\mathbf{U} = \mathbf{G}^{U,Z} d\mathbf{Z} = -\mathbf{H}_\mathbf{U}^{-1}\mathbf{H}_\mathbf{Z} d\mathbf{Z}, \quad d\mathbf{Y} = \mathbf{G}^{Y,Z} d\mathbf{Z} = \mathbf{Y}_U d\mathbf{U} + \mathbf{Y}_Z d\mathbf{Z}$$

- $d\tilde{\mathbf{X}}_t$ is the stochastic process we care about (with consumption, investment, etc), and has autocovariances

$$\text{Cov}\left(d\tilde{\mathbf{X}}_t, d\tilde{\mathbf{X}}_{t'}\right) = \sum_{s=0}^{T-1-(t'-t)} \left(d\mathbf{X}_s\right)\left(d\mathbf{X}_{s+t'-t}\right)'$$

- That only depends on the distance $t' - t$ and not time directly.

- In Krusell Smith, there's a single shock $Z$,
  - with implied autocorralations given by the $\rho Z_{t-1}$iterated term, which decay exponentially
  - More generally, for any outcome, the autocorrelation Corr $(dX_{t+l}, dZ_t)$ shows that capital and consumption—and to a much lesser extent, output—tend to lag productivity.

- These autocovariances can be stacked to form $\mathbf{V}$ and can be used directly to *calibrate or estimate* a model—as in the SMM method, but without the need for explicit simulation.
- They can be used to evaluate the likelihood function, as an input to likelihood-based estimation on time series data.
- **Evaluating the Likelihood Function**
- Let the $n_{obs}$ sized vector of observables whose likelihood we would like to determine be

$$d\tilde{\mathbf{X}}_t^{obs} = Bd\tilde{\mathbf{X}}_t + \mathbf{u}_t$$

- Where $d\tilde{\mathbf{X}}_t = \left(d\tilde{X}_t^o\right)_{o\in\mathcal{O}}$ is again the vector of stochatic processes of all $n_o$ outputs
- Recall $d\tilde{X}_t^o = \sum_{s=0}^{T-1}\sum_{z\in\mathcal{Z}} dX_s^{o,z}\epsilon_{t-s}^z, \quad d\mathbf{X}^{o,z} = \mathbf{G}^{o,z}d\mathbf{Z}^z, \quad d\tilde{Z}_t^z = \sum_{s=0}^{\infty} dZ_s^z\epsilon_{t-s}^z$
- $\{\mathbf{u}_t\}$ is i.i.d. $N\left(0,\Sigma_u\right)$, measurement errors. Will actually use $\Sigma_{\mathbf{u}} = 0$ as a prior.
- One can think about trying to predict the empirical observables $d\tilde{\mathbf{X}}_t^{obs}$ using the stochatic processes for all $n_o$ $\tilde{\mathbf{X}}_t$.
- $B$ is a $n_{obs}\times n_o$ matrix – these are coefficients in a linear regression. Naturally, we'd like to minimize the sum of square residuals if we were doing least squares, or maximize the likelihood in MLE.
- $d\tilde{\mathbf{X}}_t^{obs}$ is thus a linear combination of $\epsilon_t^z$ and $\mathbf{u}_t$, and has a multivariate normal distribution.
- Since it's just a linear transformation of $\tilde{\mathbf{X}}_t$ and $\mathbf{u}_t$, it's second moments are as well

$$\text{Cov}\left(d\tilde{\mathbf{X}}_t^{obs}, d\tilde{\mathbf{X}}_{t'}^{obs}\right) = B\,\text{Cov}\left(d\tilde{\mathbf{X}}_t, d\tilde{\mathbf{X}}_{t'}\right)B' + 1_{t=t'}\cdot\Sigma_{\mathbf{u}}$$

- Where $\text{Cov}\left(d\tilde{\mathbf{X}}_t, d\tilde{\mathbf{X}}_{t'}\right) = \sum_{s=0}^{T-1-(t'-t)}\left(d\mathbf{X}_s\right)\left(d\mathbf{X}_{s+t'-t}\right)'$
- $\mathbf{V}$ is a large $n_{obs}T_{obs}\times n_{obs}T_{obs}$ symmetric matrix with covariances $Cov\left(d\tilde{\mathbf{X}}_t^{obs}, d\tilde{\mathbf{X}}_{t'}^{obs}\right)$ stacked where $T_{obs}$ is the number of periods in the data.
- Stacking the data $d\tilde{\mathbf{X}}^{obs} = \left(d\tilde{\mathbf{X}}_t^{obs}\right)$ as a $n_{obs}T_{obs}$ dimension vector, the likelihood of the data can be expressed as a function of the observed data

$$\mathcal{L} = -\frac{1}{2}\log\det\mathbf{V} - \frac{1}{2}\left[d\tilde{\mathbf{X}}^{obs}\right]'\mathbf{V}^{-1}\left[d\tilde{\mathbf{X}}^{obs}\right]$$

- **Likelihood-based estimation**
- This involves computing the likelihood function many times for different parameters $\theta$
- Using this likelihood they estimate the posterior mode, and then use a standard Markov chain Monte Carlo method (Random Walk Metropolis Hastings, RWMH) to trace out the shape of the posterior distribution

- They simply need to compute the var-covar matrix of model observations $\mathbf{V}(\theta)$ for each parameter draw.

- To compute $\mathbf{V}(\theta)$, they have to compute the IRFs each time $d\mathbf{X}(\theta)$, andn build $\text{Cov}\left(d\tilde{\mathbf{X}}_t^{obs}, d\tilde{\mathbf{X}}_{t'}^{obs}\right) = B\,\text{Cov}\left(d\tilde{\mathbf{X}}_t, d\tilde{\mathbf{X}}_{t'}\right)B' + 1_{t=t'} \cdot \Sigma_{\mathbf{u}}$ where $\text{Cov}\left(d\tilde{\mathbf{X}}_t, d\tilde{\mathbf{X}}_{t'}\right) = \sum_{s=0}^{T-1-(t'-t)} \left(d\mathbf{X}_s\right)\left(d\mathbf{X}_{s+t'-t}\right)'$

- They re-compute $d\mathbf{X}$ very efficiently since they can reuse some Jacobians.

- Consider first the estimation of the parameters of shock processes. The matrix $\mathbf{G}^{o,z}$ needed for the IRF....

$$d\mathbf{X}^{0,z} = \mathbf{G}^{o,z} d\mathbf{Z}^{z}$$

- ... is unchanged across parameter draws, since these parameters only change the vector of shocks $d\mathbf{Z}^z$, but not $\mathbf{G}^{o,z}$ to a first order.

- Thus, to estimate the parameters in the shock process it is sufficient to compute all matrices $\mathbf{G}$ only once, and the for each $\theta$ form $d\mathbf{X}$, $\mathbf{V}$ and $\mathcal{L}$ without having to solve the model again.

- Next, consider the estimation of parameters that ***do not change the steady state*** of the heterogeneous-agent problem (price stickiness, capital adjustment costs, or monetary policy rules parameters that are turned off in the steady state). These parameters also do not affect the Jacobian of the heterogeneous-agent block, since they are "simple" blocks. Thus, these Jacobians can be held fixed always $\mathbf{J}^{o,i} = \sum_{m \in \mathcal{I}_b} \mathcal{J}^{o,m} \mathbf{J}^{m,i}$ at their initial values.

- Estimating parameters that ***do change the steady state*** of the heterogeneous agent problem, then the Jacobian of that problem needs to be recomputed for each new draw of parameters, together with the new steady state.

- Re-evaluating the steady state is so costly that they can't do it yet.

- Instead, they calibrate the steady-state micro parameters governing the heterogeneous-agent problem using cross-sectional micro data anyways, which is a good practice, and use time series data only to estimate macro parameters

- **Priors**
  - assume that the priors for the standard deviations of all shocks $\sigma$ are Inverse-Gamma distributed with mean 0.4 and standard deviation 4.
  - assume that the priors for persistence parameters $\rho$ are Beta distributed with mean 0.5 and standard deviation 0.2.
  - assume no measurement error $\Sigma_{\mathbf{u}} = 0$

- **Posterior.** They search for the posterior mode using a standard optimization routine, starting with the prior mode as our initial guess and run a RWMH in which the proposal distribution is a multivariate normal with variance equal to the inverse Hessian at the posterior mode, scaled by a factor c that is adjusted to get an acceptance rate around 25%
  - They use SciPy implementation of L-BFGS-B, imposing some non-binding bounds to guide the routine away from poorly-behaved regions of the parameter space.
- **Data.** For each model, we use the same U.S. data series as those used in Smets and Wouters (2007), over the same sample period (1966:1–2004:4). We linearly de-trend the logs of all growing vari- ables (output, consumption, investment, wages, hours) and take out the sample means of inflation and nominal interest rates.
- Krusell-Smith
  - single shock TFP
  - match single time series $\left\{dX_t^{obs}\right\}$ of aggregate output $Y_t$
  - assume shock follows $ARMA(1,1)$ $(1-\rho L)d\tilde{Z}_t = (1-\theta L)\sigma\epsilon_t$ where where L denotes the lag operator
  - estimate three parameters: $\rho, \theta, \sigma$ (only about shocks), non for the model (steady state parameters can be set to match cross-section)
- One asset HANK
  - three shocks (monetary policy shocks, government spending shocks, and price markup shocks)
  - three time series to match (output, inflation, and nominal interest rates)
  - each shock is $AR(1)$ with standard deviation and persistance, yielding 6 shock parameters
  - they then do estimated shock and model parameters in the joint estimation, $\varphi$ Taylor rule, $\phi_y$ output, and sticky prices $\kappa$ for the Phyllips curve
- Two Asset HANK
  - seven shocks from Smets and Wouters (2007) to the two- asset model: shocks to TFP, government spending, monetary policy, price and wage markups.
  - estimate the parameters of these seven shock processes using time series data on output, consumption, investment, hours, wages, nominal interest rates and price inflation.
  - model params include $\kappa^p, \kappa^w$, the degree of capital adjustment costs $\epsilon_I$

Estimation via SSJ

1. **Steady State.** Calibrate steady state to match cross-sectional data, get GE Jacobians (composing the PE Jacobians along the DAG).

2. **Single IRF via GE Jacobians.** Build shock paths $d\mathbf{Z}^z$. With Jacobians $\mathbf{G}^{o,z}$, we can get the IRFs of any output $o$ to any shock $z$, and stack IRFs to make a MA representation

$$d\mathbf{X}^{o,z} = \mathbf{G}^{o,z} d\mathbf{Z}^z \to d\mathbf{X} := [d\mathbf{X}^{o,z}]_{\forall o,z}$$

3. **Time path via MA representation.** Our potentially observed outcome is a (linear) combination of different shocks each with different IRFs, and we can write the path of output $o$ as an $MA(\infty)$ or $MA(T-1)$ process that changes with *all* shocks $z \in \mathcal{Z}$ that we have in the model as what's determinend by the impulse response $d\mathbf{X}^{o,z}$

$$d\tilde{X}_t^o = \sum_{s=0}^{T-1} \sum_{z \in \mathcal{Z}} dX_s^{o,z} \epsilon_{t-s}^z$$

4. **Variance-Covariance Matrix via Second Moments.** All we nneed second moments of the shocks to fully characterize the second moments of endogenous outcomes. Since $d\tilde{X}_t^o$ is a stochastic process, we can calculate its variance-covariance structure

$$\text{Cov}\left(d\tilde{\mathbf{X}}_t, d\tilde{\mathbf{X}}_{t'}\right) = \sum_{s=0}^{T-1-(t'-t)} \left(d\mathbf{X}_s\right)\left(d\mathbf{X}_{s+t'-t}\right)'$$

5. **Evaluating the Log-Likelihood Function with Data and Models VC Matrix.** Assume that our structural model is right, and that the observed data is driven by it and some Normal i.i.d. measurement error linenarly. Then, the variance covariance structure in the data must be just a linear combo of the variance covariance structure implied by the model, and can be written as $\mathbf{V}, n_{obs}T_{obs} \times n_{obs}T_{obs}$. Stacking the data $d\tilde{\mathbf{X}}^{obs} = \left(d\tilde{\mathbf{X}}_t^{obs}\right)$ as a $n_{obs}T_{obs}$ dimension vector, the likelihood of the data can be expressed as a function of the observed data

$$d\tilde{\mathbf{X}}_t^{obs} = Bd\tilde{\mathbf{X}}_t + \mathbf{u}_t, \quad \text{Cov}\left(d\tilde{\mathbf{X}}_t^{obs}, d\tilde{\mathbf{X}}_{t'}^{obs}\right) = B\,\text{Cov}\left(d\tilde{\mathbf{X}}_t, d\tilde{\mathbf{X}}_{t'}\right)B' + 1_{t=t'} \cdot \Sigma_{\mathbf{u}}$$

$$\mathcal{L} = -\frac{1}{2}\log\det\mathbf{V} - \frac{1}{2}\left[d\tilde{\mathbf{X}}^{obs}\right]'\mathbf{V}^{-1}\left[d\tilde{\mathbf{X}}^{obs}\right]$$

6. **Optimizing the Log-Likelihood Function** (Metropolis-Hastings) to find $\theta$ evaluating $\mathbf{V}(\theta)$ every time.

**Krusell Smith example.**

- Suppose TFP shocks $dZ$ in Krusell Smith have a relatively *persistent* (but not permanent) and a *transitory* (purely i.i.d.) component

$$dZ_t = dZ_t^1 + dZ_t^2 \begin{cases} dZ_t^1 = \rho dZ_{t-1}^1 + \epsilon_t^1 \\ dZ_t^2 = \epsilon_t^2 \end{cases} \qquad \epsilon^1 \text{ and } \epsilon^2 \sim N\left(0, \sigma_1^2 \text{ and } \sigma_2^2\right)$$

- We have three outcomes of interest $o \in \{Y, C, K\}$ and these two shocks $z = 1, 2$.

- Let $m^{x,1}$ and $m^{x,2}$ be the (truncated) IRFs of variable $X$ to these two structural shocks. This is what we get from $d\mathbf{X}^{o,z} = \mathbf{G}^{o,z} d\mathbf{Z}^z$. Then, the $MA(T-1)$ representation of the outcome path adds the effects of the two shocks.

$$Y_t = \sum_{z=1}^{2} \sum_{s=0}^{T-1} m_s^{y,z} \epsilon_{t-s}^z$$

**Step 1.** Stacked impulse responses: $d\mathbf{X}_s = [d\mathbf{X}^{o,z}]$ which will be a $T, n_o, n_z$ array.

- For any two outcomes (e.g. Consumption or Investment and Output) $Y$ and $X$, their covariance takes the following form

$$\text{Cov}\left(Y_t, X_{t+l}\right) = Cov\left(Y_t, X_{t+l} \middle| \{\sigma_z\}_z\right) = \mathbb{E}\left[Y_t X_{t+l}\right] = \sum_{z=1}^{2} \sum_{s=0}^{T-1} m_s^{x,z} \mathbb{E}\left[\epsilon^z \epsilon^{z\prime}\right] m_{s+l}^{y,z} = \sum_{z=1}^{2} \sigma_z^2 \sum_{s=0}^{T-1} m_s^{x,z} m_{s+l}^{y,z}$$

**Step 2.** Obtain covariance at all leads and lags via Fast Fourier Transform $\Sigma_{T \times n_o \times n_o}$ with an entry being the covariance for any $t$ between output $o_1$ at $t$ and output $o_2$ at horizon $t+l$ with $l \in T$ on the first entry. Again, due to the Markov properties of the model, only the distance between shocks matters. This contains, for example, the covariance between $C$ and $Y$ (whether $C$ is procyclical) and whether it leads or lags output and to what extent.

**Step 3.** Stack into a matrix $\mathbf{V}(\theta)$ and give the data for all outputs $\mathbf{y}$, evaluate the log-likelihood

$$\mathcal{L} = -\frac{1}{2} \log\det \mathbf{V} - \frac{1}{2}\left[d\tilde{\mathbf{X}}^{obs}\right]' \mathbf{V}^{-1} \left[d\tilde{\mathbf{X}}^{obs}\right]$$

- aka. $\mathcal{L} = -\frac{1}{2}\log(\det(\mathbf{V})) - \frac{1}{2}\mathbf{y}'\mathbf{V}^{-1}\mathbf{y}$. We are going to evaluate this many times to get $\mathcal{L}(\theta)$ which depends on parameters via $\mathbf{V}(\theta)$.

- Note that to estimate $\rho$ and $\sigma_1/\sigma_2$ given $\sigma_2$ all we need to do is in each evaluation, re-compute the IRFs to the shocks that we are adjusting.

  - For example, if we are estimating $\rho$, we need to adjust the shock we feed to the IRF every time. For example, with a unit shock

$$dZ_t^1 = \rho dZ_{t-1}^1 + \epsilon_t^1 \rightarrow dZ_t^1 = \rho^t$$

- And then, re-compute $dX$ and the variance covariance matrix $V$ and evaluate the likelihood again.
- Fixing all parameters, if we vary only one parameter, say a $\sigma_1/\sigma_2$, we can draw the likelihood in a 2D plot, and maximize